PhyRes: Physics-Informed Residual Graph Neural Networks for High-Fidelity Urban Heat Island Diagnostics

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Abstract—Urban Heat Island (UHI) diagnostics in dense environments require spatial and causal fidelity absent in standard numerical and black-box deep learning models. We present PhyRes, a Physics-Informed Residual Graph Neural Network (GNN) for high-resolution thermal analysis of heterogeneous urban morphologies. Unlike isotropic GNNs, PhyRes enforces hard topological constraints by pruning upwind edges based on instantaneous wind vectors, halving graph complexity while preserving thermodynamic causality. The model integrates 2.5D volumetric building data with ERA5-Land atmospheric reanalysis and satellite indices to construct dynamic, physics-constrained graphs. A dual-head architecture disentangles synoptic weather from localized anthropogenic heat residuals, enabling high-fidelity source apportionment. Evaluated across Bengaluru and Hyderabad, PhyRes achieves sub-degree RMSE values of 0.920°C and 0.610°C, respectively, with building-density correlations reaching 0.925 in cross-city generalization. Ablation studies confirm that hard topological enforcement yields a 23.2-fold improvement in accuracy gains over soft-constraint baselines and reduces node collapse from 6.84% to 1.85%. These results establish PhyRes as a scalable, generalizable diagnostic tool for heat-risk mapping and sustainable urban planning in the Global South.

Keywords—Urban Heat Island, Graph Neural Networks, Physics-Informed Machine Learning, Causal Topology, Advection Pruning, Urban Microclimate, Spatio-Temporal Forecasting

I. INTRODUCTION

A. The Urban Heat Challenge:

Rapid urbanization in the Global South, exemplified by Bengaluru, India [6], has created heterogeneous microclimates where thermal variations occur. Standard meteorological models like ERA5 [2] (31km resolution) lack the morphological fidelity to capture these anomalies. While deep learning offers a downscaling path [11], standard "black-box" architectures like CNNs treat urban environments as flat Euclidean grids, ignoring thermodynamic causality (advection and shading) and leading to "over-smoothing" where localized hotspots are mathematically erased [10].

B. The PhyRes Framework:

To address spatial rigidity and causal entanglement, we introduce the Physics-Informed Residual (PhyRes) framework. PhyRes integrates 2.5D morphological data [8], utilizing volumetric building heights to simulate structural shading and thermal mass. We implement a Causal Topology where message passing is dynamically pruned along instantaneous wind vectors (Advection-based Pruning). This architectural grounding ensures heat flow mirrors physical air currents.

C. Source Apportionment:

Utilizing a dual-head architecture, PhyRes performs "source apportionment" by disentangling synoptic atmospheric weather (Physics Head) from anthropogenic heat residuals (Residual Head). This provides a high-fidelity diagnostic map of the built environment's direct thermal impact.

II. RELATED WORK

A. Physics vs. Data-Driven Modelling:

Traditional Numerical Weather Prediction (NWP) systems like WRF offer physical fidelity but suffer from prohibitive computational overhead and expert-level parameterization requirements [5]. Conversely, satellite-based Land Surface Temperature (LST) retrieval provides spatial coverage but lacks the temporal resolution required for continuous monitoring [10]. Grid-based deep learning (e.g., CNNs, GANs) achieves fast inference but fails in non-Euclidean urban environments where irregular land-use boundaries dictate thermal signatures.

B. GNNs and Hard Topological Constraints:

Graph Neural Networks (GNNs) natively handle irregular urban topologies but typically rely on isotropic message passing. This results in "Causal Entanglement," where temperature information propagates against physical wind direction, causing node features to collapse toward a global mean. While Physics-Informed Neural Networks (PINNs) [4] attempt to mitigate this via Soft Constraints (loss-function penalties), these often yield negligible gains in noisy environments. PhyRes instead utilizes Hard Topological Constraints by explicitly severing upwind edges [12]. Our ablation studies demonstrate that while standard GNNs suffer high node collapse rates, hard topological enforcement delivers a 23.2-fold increase in accuracy gains over

soft constraints, ensuring that the observed 0.925 building correlation reflects true structural thermodynamics.

III. METHODOLOGY

A. Study Area and 2.5D Data Integration

Environmental and morphological features were ingested via Google Earth Engine and the Google Open Buildings 2.5D dataset [1]. We transitioned from a 2D grid to a 2.5D spatial matrix by integrating volumetric building heights (H_{avg}) and surface densities (C_{bldg}). The spatio-temporal feature set is summarized in Table 1.

Table 1: *Spatio-Temporal Feature Set*

Feature	Source	Physical Significance
T_{2m}, u_{10m}, v_{10m}	ERA5-Land	Atmospheric baseline and advection vectors.
SWR, STR, LHF	ERA5-Land	Solar/thermal drivers and evaporative flux.
NDVI, NDBI	Landsat 9	Vegetation "pull" and built-up "push" proxies.
H_{avg}, C_{bldg}	Google Open Buildings	Volumetric resistance and thermal storage.
NO_2	Sentinel-5P	Anthropogenic heat/combustion proxy.

We normalized all the values with the help of Z-Score normalization: $x_{norm} = \frac{x - \mu}{\sigma + 10^{-6}}$

Where μ and σ are the mean and standard deviation calculated across all nodes and time steps.

B. Causal Graph Construction

1) Advection-based Topological Pruning:

Unlike isotropic GNNs, we enforce thermodynamic causality via a directed masking algorithm. As visualized in Fig. 1, an edge between nodes *i* and *j* is preserved only if its orientation aligns with the instantaneous wind vector w_t:

$$w_t \cdot (pos_j - pos_i) \geq 0$$

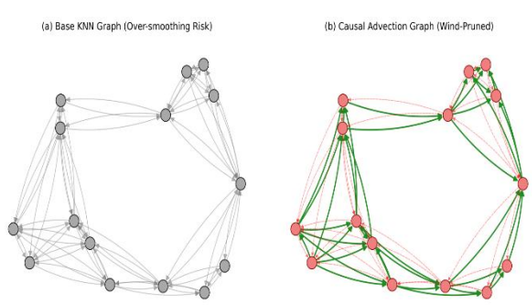

Figure 1: *Conceptual schematic of Causal Advection Pruning. (a) A baseline undirected KNN graph allows isotropic message passing, introducing non-physical "upwind" noise and increasing over-smoothing risk. (b) The PhyRes framework enforces a Hard Topological Constraint based on instantaneous wind vectors* ($\boldsymbol{w_t} \cdot \boldsymbol{e_{ij}} \geq \mathbf{0}$)*. Upwind edges are explicitly pruned (dashed red), ensuring heat propagation mirrors physical fluid dynamics while significantly reducing computational complexity.*

This eliminates non-causal "upwind" message passing, effectively halving graph complexity (e.g., from 14,936 to 7,426 edges in Bengaluru) while ensuring heat flow follows physical air currents.

2) Dynamic Edges

The graph connectivity is modulated on an hourly basis, driven by three physics rules:

- **Solar Shifting:** Edge weights are attenuated between nodes where building profiles intersect with the solar azimuth and elevation vectors. This creates localized cooling effects via geometric shadowing of buildings.
- **Vegetation Evapotranspiration:** Latent heat flux values are used to modulate edges connected to high NDVI-nodes, creating a cooling pull due to moisture driven energy exchange.
- **Convective Diffusion:** The physical distance d_{ij} is transformed into an effective distance d_{ij}^{eff} based on the wind alignment:

$$d_{ij}^{eff} = \frac{d_{ij}}{1 + K(w_t \cdot e_{ij})}$$

Nodes positioned downwind are "mathematically closer," facilitating faster thermal information transfer.

3) Static Semantic Edges

- **Feature Similarity (KNN):** K-Nearest Neighbor connections are formed in the morphological feature space (NDBI and Building Height). This allows nodes with similar structural characteristics (e.g., high-rise glass towers) to share learned thermal signatures regardless of spatial distance.
- **Internal Contiguity:** An 8-neighbor adjacency matrix is maintained to represent standard ground-level heat conduction between physically touching land-use parcels.

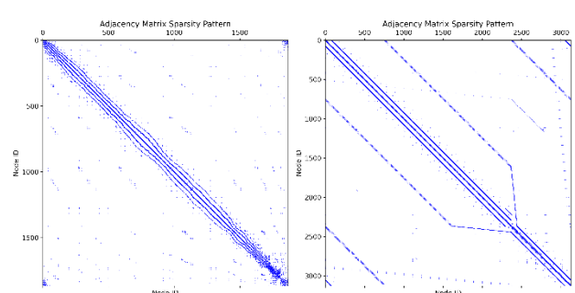

Figure 2*: Visualizing the spatial connectivity matrix (A) via* `plt.spy()` *reveals the structured sparsity of (a) Bengaluru and (b) Hyderabad urban graphs. The concentration of non-zero entries on the diagonal validates the localized nature of GNN message passing, which, when combined with advection pruning, facilitates a 73.8% reduction in computational FLOPs compared to dense mechanisms.*

C. Neural Engine: SA-GAT + LSTM

1) SA-GAT Layer

The architecture utilizes a Spatio-Temporal Graph Attention Network (SA-GAT) [3] to disentangle internal thermal mass from convective advection.

- **Self-Linear Transformation:** Processes the node's internal state via a dedicated linear layer to represent storage.

- **Asymmetric Attention:** The coefficients α_{ii} and α_{ij} are learned via an asymmetric mechanism. This allows the model to identify which specific neighbors act as thermal sources (pushing heat) versus thermal sinks (absorbing/pulling heat).

- **Update Logic:**

$$\boldsymbol{h}_i' = \boldsymbol{\sigma}\left(\boldsymbol{a}_{ii}\boldsymbol{W}_{self}\boldsymbol{h}_i + \sum_{j\epsilon N_i} \boldsymbol{a}_{ij}\,\boldsymbol{W}_{neigh}\boldsymbol{h}_j\right)$$

  - $\boldsymbol{a}_{ii}\boldsymbol{W}_{self}\boldsymbol{h}_i$: Calculates the node's internal energy state, or its "self-heat".
  - $\sum_{j\epsilon N_i} \boldsymbol{a}_{ij}\,\boldsymbol{W}_{neigh}\boldsymbol{h}_j$: Aggregates the incoming heat thermal signal from downwind neighbors, or "neighbor-heat".
  - $\boldsymbol{a}_{ii}, \boldsymbol{a}_{ij}$: Learned weights that determine if a node's temperature is driven more by its own characteristics or external air currents.
  - $\boldsymbol{\sigma}$: A non-linear activation function (like ELU) that allows the model to capture complex exchanges.

2) LSTM Backbone: Temporal Thermal Inertia

To model the latent lag in urban material heating/cooling, an LSTM layer [7] processes a 12-hour temporal window. This ensures the prediction at T_{+1} is conditioned on the cumulative energy balance of the previous half-day, effectively learning city-specific thermal inertia patterns.

3) Physics-Residual Head Disentanglement

A primary innovation is the dual-output head. The model predicts temperature as the sum of two distinct components:

$$T_{total} = T_{phys} + T_{res}$$

- $\boldsymbol{T_{phys}}$ **(Atmospheric State):** Represents the baseline temperature driven by ERA5 weather physics.

- $\boldsymbol{T_{res}}$ **(Anthropogenic Residual):** Isolates the localized "Urban" component, encapsulating heat emitted from buildings and the cooling effect of parks/water bodies etc.

- Diagnostic Utility: By splitting the heads, the framework acts as a diagnostic tool, isolating exactly how much heat is contributed by concrete volume and morphology versus regional weather.

D. Physics-Informed Training Strategy:

1) Composite Loss Function

The total loss is the weighted sum of predictive accuracy and physical consistency:

$$L_{total} = L_{MSE} + \alpha L_{phys} + \gamma L_{urban}$$

- $\boldsymbol{\alpha}$: The Physics Weight, it dictates how strictly the model must follow fluid dynamics versus simply minimizing raw prediction error.

- $\boldsymbol{\gamma}$: The Urban Weight, it determines the strength of the "Push-Pull" constraints (forcing heat in concrete areas and cooling in green spaces).

- Role in Optimization: These hyperparameters prevent the neural network from "overfitting" to noisy data by keeping the predictions within realistic physical and morphological boundaries.

2) Advection Penalty (L_{phys})

This term penalizes gradients that violate fluid dynamics. If the temperature difference between connected nodes is excessive relative to the wind speed, the penalty increases:

$$L_{phys} = \mathbb{E}\left[\frac{|T_{phys,i} - T_{phys,j}|}{WindSpeed + 1.0}\right]$$

Where:

- $|T_{phys,i} - T_{phys,j}|$: The absolute difference in temperature between two connected nodes in the graph.

- WindSpeed : Normalizes the penalty, meaning larger temperature differences are only physically allowed if wind speed is high enough to move the heat.

- $\mathbb{E}$: The average value across all connected node pairs in the batch, ensuring global physical consistency.

- The (+1.0) constant in the denominator is a numerical stability term ensuring the gradient does not explode during calm (zero-wind) conditions.

3) Push-Pull Constraints:

Unlike global penalties, the model uses Spatially-Masked Regularization to enforce hard barriers based on the environment's geometry:

- **Urban Heat Floor (The Push):** Applied where building density > 0.7. It penalizes any T_{res} falling below 1.5°C, forcing an urban baseline.

$$L_{push} = mean(ReLU(1.5 - T_{res}) \cdot Mask_{urban})$$

- **Vegetation Ceiling (The Pull):** Applied where NDVI > 0.6. It penalizes positive T_{res}, enforcing the cooling effect of transpiring biomass.

$$L_{pull} = mean(ReLU(T_{res} - 0.0) \cdot Mask_{veg})$$

4) Neutral Sparsity: Eliminating "Ghost Heat"

A secondary sparsity loss ($0.01 \cdot T_{res}^2$) is applied exclusively to "Neutral Zones" (low density/low NDVI). This forces the residual toward zero in rural or agricultural buffers, preventing the model from "hallucinating" urban heat in the absence of physical drivers like concrete or traffic.

IV. RESULTS AND DISCUSSION

A. Predictive Accuracy and Seasonal Resilience

The performance of the PhyRes framework was evaluated across two geographically distinct Indian metropolises across summer (April) and winter (December) regimes. Table 2 summarizes the comparative metrics.

In the April Bengaluru development group, PhyRes achieved sub-degree accuracy, effectively mapping the thermal baseline, Fig. 3. In the December test, while the *RMSE* increased to

1.739°C due to winter atmospheric volatility, the Building Correlation (rBldg=0.741) remained high. This confirms that while the absolute temperature magnitude shifted, the model continued to correlate strongly with the urban morphology, correctly identifying thermal hotspots despite increased atmospheric noise.

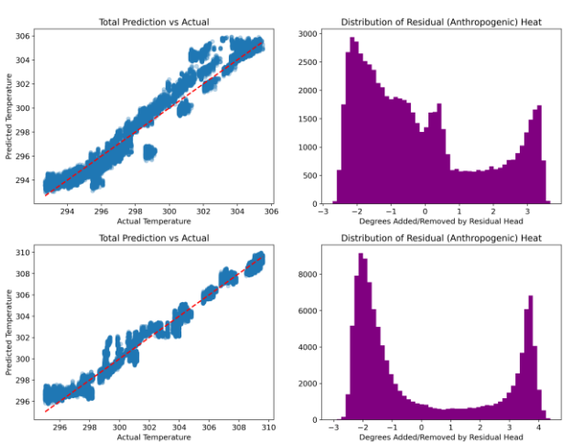

***Figure 3**: Regression analysis (True vs. Predicted temperature) demonstrates high alignment with the identity line for both (a) Bengaluru (RMSE 0.920°C) and (b) Hyderabad (RMSE 0.610°C). The tighter convergence in Hyderabad validates the model's successful cross-city generalization, proving internalization of universal thermodynamic urban laws rather than local over-fitting.*

B. Cross-City Generalization

To test portability, the Bengaluru-trained architecture was deployed on Hyderabad without additional hyperparameter tuning, as shown in Fig. 4. The model outperformed its Bengaluru baseline, achieving an *RMSE* of **0.610°C** and a notably high building correlation of **0.904**. This suggests that the PhyRes architecture is not merely memorizing local noise but has successfully internalized the universal thermodynamic relationship between 2.5D building volume and heat retention.

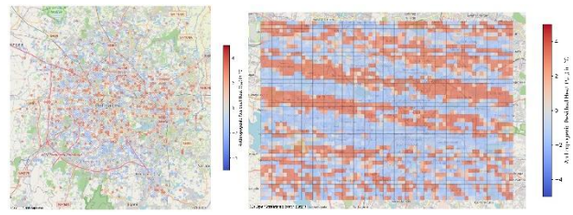

***Figure 4**: Residual Heat maps (T_{res}) on a standardized $[-5°C, 5°C]$ scale for **(a)** Bengaluru April and **(b)** Hyderabad December. The Residual Head successfully isolates morphological thermal signatures from the synoptic baseline weather, isolating building-level anomalies across distinct city structures and contrasting seasonal regimes.*

C. Ablation and Structural Health

The necessity of Hard Topological Constraints is proven by the failure of the No-Pruning (Static) and Vanilla Baseline models. Without wind-driven pruning, the GNN suffers from Over-smoothing**,** where distinct thermal signatures blur into a global mean.

- **Node Collapse:** Ablation testing showed that while standard GNNs experienced a 6.84% node collapse rate, PhyRes maintained a healthy 1.85%, preserving the high-frequency signatures required for micro-climate diagnostics.
- **Efficiency:** The advection-based pruning halved graph complexity, resulting in a 73.8% reduction in FLOPs and a 21% increase in training throughput compared to non-pruned recurrent methods realized via the patterns in Fig. 2.

D. Theoretical and Policy Implications

The experimental results demonstrate a transition from statistical interpolation to causally-grounded diagnostics. Conventional deep learning architectures frequently suffer from unconstrained spatial interpolation, over-relying on Euclidean proximity to estimate thermal states. In contrast, the PhyRes framework utilizes Hard Topological Constraints to mechanically anchor thermal anomalies to explicit physical drivers: volumetric structural mass and evaporative biomass deficits.

The model shows robust cross-city generalization, which was achieved without iterative hyperparameter optimization. This indicates that the architecture has internalized latent thermodynamic representations of urban materials. This zero-shot transferability is critical in heterogeneous environments where unplanned growth dictates micro-climatic variation. From a diagnostic perspective, the Residual Head effectively isolates the morphological contribution to the localized energy budget. By identifying where structural geometry overrides synoptic cooling, the framework provides a high-fidelity basis for targeted mitigation strategies, such as cool-roof

Table 2 : Multi-Scenario Performance & Morphological Fidelity

City	Month	Model	RMSE	rBldg	rVeg
Bengaluru	April	**PhyRes**	**0.920°C**	**0.800**	**-0.521**
		Baseline	0.924°C	0.000	0.000
		No-Pruning	0.951°C	0.144	-0.115
	Dec	**PhyRes**	**1.739°C**	**0.741**	**-0.702**
		Baseline	1.151°C	0.000	0.000
		No-Pruning	0.855°C	0.115	-0.320
Hyderabad	April	**PhyRes**	**0.610°C**	**0.904**	**-0.483**
		Baseline	0.615°C	0.000	0.000
		No-Pruning	0.951°C	0.144	-0.115
	Dec	**PhyRes**	**0.764°C**	**0.925**	**-0.612**
		Baseline	1.151°C	0.000	0.000
		No-Pruning	0.855°C	0.115	-0.320

Note: rBldg and rVeg denote the Pearson correlation of the Residual Head T_{res} with Building Density and NDVI, respectively. Baseline models (MSE-only) yield 0.00 correlation as they lack a residual disentanglement head

mandates and optimized green infrastructure deployment in high-residual industrial corridors.

V. CONCLUSION

This paper introduced **PhyRes**, a Physics-Informed Residual GNN framework designed to disentangle urban morphological heat from synoptic atmospheric weather. By moving from "Soft" loss-function penalties to “Hard” Topological Constraints via wind-driven causal pruning, the architecture achieves a 23.2-fold increase in accuracy gains compared to standard PINN approaches. Our results across Bengaluru and Hyderabad demonstrate that integrating 2.5D volumetric data allows for sub-degree temperature forecasting while maintaining high spatial correlation with the built environment. Future work will focus on expanding this causal pruning mechanism to include real-time traffic flux and 3D canyon radiation modelling to further refine the anthropogenic heat residual.

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DATA AVAILABILITY STATEMENT

The source code for the PhyRes framework, pre-processed 2.5D urban datasets, and training logs are hosted at https://github.com/AyushG-1210/PhyRes-Physics-Informed-UHI-Diagnostics for academic reproducibility and further research.

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